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Primary care experiences of veterans with opioid use disorder in the Veterans Health Administration

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ABSTRACT

Background: While patients with substance use disorders (SUDs) are thought to encounter poor primary care experiences, the perspectives of patients with opioid use disorder (OUD), specifically, are unknown. This study compares the primary care experiences of patients with OUD, other SUDs and no SUD in the Veterans Health Administration.

Methods: The sample included Veterans who responded to the national Patient-Centered Medical Home Survey of Healthcare Experiences of Patients, 2013–2015. Respondents included 3554 patients with OUD, 36,175 with other SUDs, and 756,386 with no SUD; 742 OUD-diagnosed patients received buprenorphine. Multivariable multinomial logistic regressions estimated differences in the probability of reporting positive and negative experiences (0–100 scale) for patients with OUD, compared to patients with other SUDs and no SUD, and for OUD-diagnosed patients treated versus not treated with buprenorphine.

Results: Of all domains, patients with OUD reported the least positive experiences with access (31%) and medication decision-making (35%), and the most negative experiences with self-management support (35%) and provider communication (23%). Compared to the other groups, patients diagnosed with OUD reported fewer positive and/or more negative experiences with access, communication, office staff, provider ratings, comprehensiveness, care coordination, and self-management support (adjusted risk differences[aRDs] range from |2.9| to |7.0|). Among OUD-diagnosed patients, buprenorphine was associated with more positive experiences

Abbreviations: aRD, adjusted risk difference; OUD, opioid use disorder; PCMH-SHEP, Patient-Centered Medical Home Survey of Healthcare Experiences of Patients; RAPID, Office of Reporting, Analytics, Performance, Improvement and Deployment; SUD, substance use disorder; VHA, Veterans Health Administration

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with comprehensiveness (aRD = 8.3) and self-management support (aRD = 7.1), and less negative experiences with care coordination (aRD = −4.9) and medication shared decision-making (aRD = −5.4).

Conclusions: In a national sample, patients diagnosed with OUD encounter less positive and more negative experiences than other primary care patients, including those with other SUDs. Buprenorphine treatment relates positively to experiences with care comprehensiveness, medication decisions, and care coordination. As stakeholders encourage more primary care providers to manage OUD, it will be important for healthcare systems to attend to patient access and experiences with care in these settings.

1. Introduction

The number of patients in the United States who have been diagnosed with an opioid use disorder (OUD) has grown rapidly in the last decade (Compton & Volkow, 2006; Volkow & Collins, 2017). Primary care is viewed as a promising clinical environment to engage patients, often newly diagnosed with OUD, through a longitudinal chronic care model of treatment with pharmacotherapy as an anchoring component of the care (McLellan et al., 2014; Sullivan, Chawarski, O'Connor, Schottenfeld, & Fiellin, 2005; Wakeman & Barnett, 2018). In 2002, buprenorphine and buprenorphine/naloxone (hereafter, buprenorphine) were approved in the United States for treatment of OUD in primary care and other office-based settings. Since then, the total annual number of office-based buprenorphine treatment visits has increased from 0.13 million to 2.1 million (Turner, Kruszewski, & Alexander, 2015). These national trends have been observed in the Veterans Health Administration (VHA), the largest direct provider of addiction treatment in the United States (Gellad, Good, & Shulkin, 2017; Valenstein-Mah, Hagedorn, Kay, Christopher, & Gordon, 2018; Wyse et al., 2018). While opportunities to manage OUD in primary care exist, the perspectives of patients in such settings are largely unknown.

Healthcare systems value patient perceptions of care because positive patient experiences positively correlate with patient and system-level outcomes. In general primary care populations, patients who report more positive care experiences also use more preventive services and adhere to provider recommendations (Doyle, Lennox, & Bell, 2013; Fenton, Jerant, Bertakis, & Franks, 2012; Villani & Mortensen, 2013); they are less likely than patients with fewer positive experiences to be hospitalized for ambulatory sensitive hospital conditions (Augustine, Nelson, Fihn, & Wong, 2019). These benefits may extend to substance use disorder (SUD) outcomes as well. In one prospective study of primary care patients diagnosed with alcohol, cocaine, or heroin use disorders, positive primary care experiences with access, visit continuity, and whole-person knowledge of care were associated with reductions in drug addiction severity (Kim et al., 2007).

There is limited research focused on the primary care experiences of patients with OUD, specifically. In the VHA, which routinely administers surveys of patient experience, prior studies have found that patients with any SUD reported fewer positive experiences with care than patients without any SUD in several domains, including access, provider communication, and care coordination (Burnett-Zeigler, Zivin, Ilgen, Islam, & Bohnert, 2011; Desai, Stefanovics, & Rosenheck, 2005; Hoggatt, Frayne, Saechao, Yano, & Washington, 2019). Patients with OUD may have even worse experiences with care than patients diagnosed with other SUDs. Patients with OUD have elevated rates of chronic medical conditions, psychiatric disorders, and social determinants of health (Han et al., 2017) that together contribute to fragmented care experiences. Disputes with providers over pain medications could contribute to dissatisfaction with care, and negative experiences with medication decisions and provider communication.

We compared the primary care experiences of patients diagnosed with OUD to those of patients diagnosed with other SUDs and patients without any SUD in a national sample of VHA primary care patients. We hypothesized that patients diagnosed with OUD would have fewer positive or more negative experiences with care than patients without any SUD, and those with other SUDs, in interpersonal domains (e.g., office staff helpfulness/courtesy, provider communication) and integrated care domains (e.g., care coordination, comprehensiveness, medication decision-making) in particular. Among OUD-diagnosed patients, we further explored the perspectives of patients receiving buprenorphine treatment. We focus on buprenorphine, specifically, because buprenorphine is the most common medication treatment for OUD, and national initiatives in VHA and other healthcare systems have tried to increase buprenorphine prescribing in primary care settings. While the study did not assess whether buprenorphine was prescribed in primary care or other VHA care settings, patient experiences with primary care could have implications for ongoing efforts to manage and treat OUD in primary care. Specific hypotheses regarding the relationship of buprenorphine with primary care experiences were not proposed due to the exploratory nature of those analyses.

2. Materials and methods

2.1. Design

We conducted a cross-sectional analysis of primary care experiences reported on a national survey, collected from October 2012 to September 2015.

2.2. Data sources and study sample

We used data from the 2013–2015 Patient-Centered Medical Home-Survey of Healthcare Experiences of Patients (PCMH-SHEP), an ongoing survey of VHA primary care experiences conducted by the VHA Office of Reporting, Analytics, Performance, Improvement and Deployment (RAPID). Each month RAPID identifies veteran outpatients who visited an assigned primary care provider in the past 10 months, and who did not participate in the prior year's survey. A stratified, random sample of eligible patients are mailed a letter explaining the goals of the survey, followed by the survey the second week, and a thank you/reminder postcard the third week (VHA Office of Analytics and Business Intelligence, 2012).

We linked patient responses from the PCMH-SHEP to clinical diagnoses and sociodemographic characteristics, drawn from VHA administrative records in the year prior to the survey. Patients were eligible for inclusion in analyses if they had non–missing data on study variables. Institutional review boards at the University of Utah, VA Salt Lake City Healthcare System, and VA Pittsburgh Healthcare System approved all study procedures.

2.3. Outcome variables

We examined patient experiences with care in eight domains from the PCMH-SHEP, which is based on the validated Consumer Assessment of Healthcare Providers and Systems (CAHPS) survey (Hays et al., 2014; Scholle et al., 2012; Stucky, Hays, Edelen, Gurvey, & Brown, 2016). The PCMH-SHEP asks patients to confirm that they visited a specific VA primary care provider in the past 12 months, then instructs patients to report on experiences with the specific primary care provider (see PCMH-SHEP instructions below).

Our records show that you got care from the provider named below in the last 12 months. Is that right?

The questions in this survey will refer to the provider named [above] as "this provider." Please think of that person as you answer the survey.

The following PCMH-SHEP items assess experiences in 8 domains: *access* (how often patients got timely appointments, care, and information), *communication* (how well providers communicated with patients), *office staff* (how often staff were helpful, courteous, and respectful), *comprehensiveness* (whether providers pay attention to mental health, emotions, and substance use), *support for self-management* (whether providers supported patients in taking care of their own health), *medication shared decision-making* (whether providers discussed medication decisions with patients), *care coordination* ("In the last twelve months, how often did the provider seem informed and up-to-date about the care you got from specialists?"), and *overall provider rating* (0–10, where 0 is the worst and 10 is the best provider possible).

In many domains, experience items are preceded by a screener item with instructions to skip over subsequent nonapplicable items (e.g., *In the last 12 months, did you and this provider talk about starting or stopping a prescription medicine? If no, go to #32*). For domains with multiple items, we calculated composite scores as the average of applicable, nonmissing items for each individual. We categorized the domain scores as positive, moderate, or negative using previously established cut-offs (Jones et al., 2017).

2.4. Independent variables

We determined OUD and other SUD diagnoses from VHA administrative records in the 12 months prior to the PCMH-SHEP, the same time period in which respondents were instructed to report on their experiences with primary care. Patients were coded as diagnosed if they had one inpatient visit or two outpatient visits (Frayne et al., 2010) with International Classification of Diseases version 9 (ICD-9) diagnoses for OUD (ICD-9 = 304.0, 304.7, 305.5). Other SUDs included ICD-9 diagnoses for alcohol-related disorders (ICD-9 = 291, 303, 305.0, 357.5, 425.5, 535.3, 571.0–571.3, 760.71, 980.0), and drug use disorders related to amphetamines (ICD-9 = 304.4, 305.7), cannabis (ICD-9 = 304.3, 305.2), cocaine (ICD = 304.2, 305.6), hallucinogens (ICD-9 = 304.5, 305.3), sedatives (ICD-9 = 304.1, 305.4), and other substances or unspecified combinations of drugs (ICD-9 = 304.6–304.9, 305.8, 305.9, 648.3).

In the subsample of patients diagnosed with OUD, we used VHA pharmacy records and outpatient visits to determine OUD treatment in the 12 months prior to the PCMH-SHEP. Patients were coded as receiving buprenorphine treatment if they were prescribed "buprenorphine" or "buprenorphine/naloxone" from any VHA provider; naltrexone if they were prescribed "naltrexone", and methadone if they visited a VHA- and Substance Abuse and Mental Health Services Administration (SAMSHA)–approved methadone treatment program. Because VHA methadone treatment programs dispense methadone and fill buprenorphine prescriptions, patients were coded "yes" receiving methadone treatment if they visited a treatment program without a buprenorphine or naltrexone prescription; and "no" if they did not visit a methadone treatment program, or if they visited a methadone treatment program and were also prescribed buprenorphine or naltrexone.

2.5. Covariates

We controlled for variables (Table 1) that have been shown to be associated with patient experiences with care in prior research (Burnett-Zeigler et al., 2011; Hausmann, Gao, Mor, Schaefer Jr., & Fine, 2013). From the PCMH-SHEP, we controlled for patient-reported race/ethnicity, education, general health and mental health ratings (categorized as fair/poor vs. excellent/very good/good), and number of primary care visits in the prior year (i.e., *In the last 12 months, how many times did you visit this provider to get care for yourself?*). From administrative data, we selected prior year age, sex, and marital status. We used the Gagne Comorbidity Index, a weighted index of chronic conditions predictive of one-year mortality, as a measure of medical morbidity (Gagne, Glynn, Avorn, Levin, & Schneeweiss, 2011). We controlled for non-SUD mental health disorder diagnoses, defined as one inpatient or two outpatient visits with ICD-9 diagnoses in the prior year (Frayne et al., 2010). Types of mental health disorders (e.g., mood, anxiety, etc.) were based on the Healthcare Cost and Utilization Project's Clinical Classification Software for Mental Health and Substance Abuse (Bao & Sturm, 2001), modified in the current study to include post-traumatic stress disorder. Because unstable housing is more common in patients with SUDs and is a factor in patient healthcare experiences (Jones et al., 2017), we controlled for evidence of homelessness documented in administrative records in the prior 12 months. To account for regional and facility variations in OUD treatment options and patient experiences with primary care, we controlled for patient residence (urban versus rural), facility type (medical center versus outpatient clinic), and VHA service region.

2.6. Statistical analyses

Analyses were conducted using Stata version 15 (StataCorp, 2017). We applied PCMH-SHEP survey weights to account for participant selection and survey nonresponse. Standard errors were estimated using a Taylor series approximation that accounted for the stratified sampling design. We used chi-square tests to examine overall differences in sociodemographic characteristics, based on OUD and other SUD diagnoses. We used multinomial logistic regression models to estimate differences in reporting positive, moderate, or negative experiences with care for patients with OUD, compared to patients with other SUDs and no SUD. Once the unadjusted estimates were obtained, we reran the study models controlling for sociodemographic and clinical characteristics. We converted the multinomial logistic model coefficients to predicted probabilities of reporting positive experiences and negative experiences (0 to 100 scale), and differences in predicted probabilities (risk differences) to aid interpretation. Because a large percentage of patients with other SUDs were diagnosed with alcohol use disorder, we repeated the analyses considering alcohol and drug use disorders separately.

Finally, we compared the experiences of diagnosed patients treated and not treated with buprenorphine. Specifically, we restricted the cohort to patients with OUD and repeated the multinomial logistic regression modeling procedures to estimate differences in the probability of reporting positive and negative experiences associated with buprenorphine treatment. Sensitivity analyses compared the experiences of patients treated or not treated with buprenorphine or naltrexone; then compared experiences of patients treated or not treated with any medication treatment (i.e., buprenorphine, naltrexone or methadone).

3. Results

Of the 1,989,103 patients mailed the PCMH-SHEP, 45% (n = 887,878) returned surveys, including 19% of patients with OUD, 30% of patients with other SUDs, and 46% of patients with no SUD. Patients who returned the PCMH-SHEP were more likely than nonrespondents to be age 55 or older (89% vs 63%, respectively), male (95% vs 90%), and married (62% vs 49%); they were less likely to have psychiatric diagnoses (24% vs 37%) and recent experiences of homelessness (3% vs 8%). The patterns of sociodemographic characteristics

Table 1
Sociodemographic and clinical characteristics of patients diagnosed with opioid use disorder, compared to patients diagnosed with other substance use disorders or no substance use disorder.

Characteristic	Opioid use disorder (n = 3554)	Other substance use disorders (n = 36,175)	No substance use disorder (n = 753,386)
	%	%	%
Female sex	8.8	6.0	8.2
Age			
18–44	28.0	19.4	13.1
45–54	21.6	22.6	11.9
55–64	37.9	35.8	24.1
65+	12.5	22.2	51.0
Race/ethnicity			
Non-Hispanic white	65.3	58.0	71.7
Non-Hispanic black	20.3	27.8	15.7
Hispanic	9.8	9.2	8.1
Other	4.6	5.0	4.5
Education			
< 12 years education	5.7	6.0	7.6
High school equivalent	32.3	35.0	29.9
Some college	50.7	46.2	41.1
4 year college degree	11.4	12.8	21.4
Marital status			
Married	27.3	32.9	58.2
Single, never married	23.1	21.7	10.9
Separated, divorced, widowed	49.1	44.7	29.0
Rural residence	20.7	26.5	35.8
Homelessness	33.9	25.3	3.4
General health fair/poor	53.6	48.0	37.7
Mental health fair/poor	59.2	57.3	29.7
Gagne comorbidity (mean)	1.5	1.7	0.5
Psychiatric diagnoses			
Mood disorder	65.6	57.0	17.9
Post-traumatic stress disorder	40.0	36.2	12.9
Other anxiety disorders	27.9	21.5	7.4
Psychotic disorder	9.6	7.9	1.9
Substance use disorder diagnoses			
Alcohol use disorder	43.7	83.0	
Cannabis use disorder	21.4	18.3	
Cocaine use disorder	27.1	15.4	
Hallucinogen use disorder	0.8	0.1	
Sedative use disorder	7.8	1.0	
Other/unspecified combination	32.0	16.2	
OUD treatment			
Buprenorphine	25.0		
Naltrexone	3.3		
Methadone	15.7		
None of the above	56.6		
Number of primary care visits			
One or two	45.0	47.4	60.5
Three or four	36.1	34.4	28.6
Five or more	18.9	18.3	11.0

Source: Respondents to the Veterans Health Administration's Patient-Centered Medical Home Survey of Healthcare Experiences of Patients (PCMH-SHEP), 2013–2015. % - survey weighted percent of patients with sociodemographic and clinical characteristics. Standard error of each estimate was < 1.5 for patients with opioid use disorder, and < 1.0 for other groups. Survey design adjusted X^2 tests of differences were used to assess overall group differences in sociodemographic and clinical variables. All tests were statistically significant at $p < .001$.

associated with survey response were similarly observed among subgroups of patients diagnosed with OUD, other SUDs, and no SUD (Appendix B). After excluding records for 70,085 (8%) patients with missing data on study variables, the final sample included 3554 patients with OUD, 36,175 with other SUDs, and 756,386 with no SUD. Among patients diagnosed with OUD, 25% received buprenorphine, 3% naltrexone, and 16% methadone; while 57% received no medication treatment for OUD.

3.1. Primary care experiences of patients with OUD compared to patients with other SUDs and no SUD

Only half of patients with OUD reported positive care experiences in any single domain (Table 2), with the fewest positive experiences with access (31%) and medication shared decision-making (38%); and the most positive experiences with comprehensiveness (53%) and communication (50%). More than one in seven patients with OUD reported negative experiences in any single domain, with the most negative experiences with self-management support (35%) and communication (23%), and the fewest negative experiences with care coordination (15%) and access (18%).

The X^2 tests revealed significant variations in patient experiences by SUD diagnosis (all p-values < .001). Patients with OUD reported fewer positive or more negative experiences with care than patients with other SUDs and those without any SUD in 7 domains: access, communication, office staff, overall provider rating, coordination, medication shared decision-making, and support for self-management (Table 2 and Appendix C). In the domain of comprehensiveness, patients with OUD reported more positive experiences, when compared to patients without any SUD.

Among patients diagnosed with OUD, none of the X^2 tests of differences for patients with buprenorphine, compared to patients without

Table 2
Percent of VHA patients with opioid use disorder who reported positive, moderate, and negative primary care experiences, compared to patients with other substance use disorders or no substance use disorder.

Domain (N)	Opioid use disorder (n = 3554)	Other substance use disorders (n = 36,175)	No substance use disorder (n = 753,386)	X^2
	%	%	%	
Access (*n* = 680,047)				105.6
Positive	31.2	36.5	43.4	
Moderate	51.1	51.6	47.9	
Negative	17.7	11.9	8.7	
Communication (*n* = 791,642)				160.9
Positive	49.8	55.5	62.8	
Moderate	26.8	26.9	25.4	
Negative	23.4	17.6	11.9	
Office staff (*n* = 792,218)				62.4
Positive	46.6	56.1	59.0	
Moderate	32.2	29.4	29.4	
Negative	21.1	14.5	11.6	
Overall provider rating (*n* = 790,141)				110.1
Positive	39.5	43.5	48.2	
Moderate	39.9	41.8	41.5	
Negative	20.6	14.7	10.3	
Comprehensiveness (n = 792,665)				260.7
Positive	52.8	56.3	42.6	
Moderate	28.1	28.2	37.8	
Negative	19.1	15.5	19.6	
Coordination (*n* = 504,535)				25.5
Positive	48.1	54.4	56.5	
Moderate	36.8	33.8	34.0	
Negative	15.1	11.8	9.5	
Medication shared decision-making (*n* = 481,117)				18.9
Positive	38.2	40.2	39.3	
Moderate	43.6	44.7	47.8	
Negative	18.1	15.0	12.9	
Support for self-management (*n* = 789,746)				19.8
Positive	42.8	47.6	44.2	
Moderate	22.5	24.9	26.0	
Negative	34.7	27.5	29.8	

Source: Respondents to the Veterans Health Administration's Patient-Centered Medical Home Survey of Healthcare Experiences of Patients (PCMH-SHEP), 2013–2015. N – number of PCMH-SHEP respondents with 1+ applicable, non-missing item in this domain. % - survey weighted percent of patients with positive, moderate, or negative care experiences. Standard errors of each estimate was < 1.5 for patients with opioid use disorder, and < 1.0 for other groups. Survey design adjusted X^2 tests of differences were used to assess overall group differences in reporting positive, moderate, and negative experiences with care. All tests were statistically significant at $p < .001$.

buprenorphine, reached statistical significance (all p-values > .05; not tabled).

3.2. Adjusted differences in the probability of reporting positive and negative care experiences for patients with OUD compared to patients with other SUDs and no SUD

While differences in primary care experiences between patients with OUD and patients with other SUDs were somewhat smaller after controlling for patient sociodemographic and clinical characteristics, most differences remained statistically significant (Table 3). Compared to patients with other SUDs, patients diagnosed with OUD reported fewer positive and more negative experiences in five domains: access, communication, office staff, comprehensiveness, and self-management support (adjusted risk differences [aRDs] for positive experiences = −3.2 to −7.0; aRDs for negative experiences = 2.9 to 6.2). They reported fewer positive experiences with care coordination (aRD = −4.3) and more negative provider ratings (aRD = 3.2).

Compared to patients without any SUD, patients diagnosed with OUD reported fewer positive and more negative experiences with care in five domains: communication, office staff, provider ratings, care coordination, and self-management support (aRDs for negative experiences = −4.3 to −5.5; aRDs for positive experiences = 3.5 to 5.4). They also reported more negative experiences with access (aRD = 4.0).

3.3. Differences in the probability of reporting positive experiences for patients with OUD compared to patients with alcohol use disorders (AUDs) and other nonopioid drug use disorders

Among patients with other SUDs, 24,264 (67%) had diagnoses of AUD only, while 11,911 (33%) had diagnoses of drug use disorders excluding OUD. Adjusted probabilities of reporting positive and negative experiences with care for patients with OUD, other drug use disorders, AUD, or no SUD are provided in Appendix D (Table D.1 and Table D.2). Compared to patients with other drug use disorder diagnoses, those with OUD reported fewer positive and more negative experiences with office staff (aRDs = −5.8 and 3.2, respectively), and more negative experiences with access (aRD = 2.8). Compared to patients with AUD only, patients with OUD reported fewer positive and/or more negative experiences in all 8 domains (aRDs for positive experiences = −3.0 to −7.8; aRDs for negative experiences = 2.8 to 8.6).

3.4. Primary care experiences of patients with OUD treated or not treated with buprenorphine

In the sample of respondents diagnosed with OUD, some differences in primary care experiences between patients treated or not treated with buprenorphine became statistically significant in the adjusted models (Table 4). After controlling for observed group differences in age, race, and other sociodemographic and clinical covariates, patients with buprenorphine were more likely than patients without

Table 3
Adjusted differences in positive and negative primary care experiences for patients diagnosed with opioid use disorder, compared to patients with other substance use disorders or no substance use disorder.

Primary care domain	Probability of positive experience			Comparison: OUD minus other SUD		Comparison: OUD minus no SUD	
	OUD	Other SUD	No SUD	aRD	95% CI	aRD	95% CI
Access	39.6	42.8	42.9	-3.2*	−6.0, −0.4	−3.3*	−5.9, −0.6
Communication	57.5	61.0	62.4	−3.5**	−6.1, −0.9	−4.9***	−7.4, −2.4
Office Staff	53.2	60.1	58.6	−7.0***	−9.6, −4.4	−5.5***	−8.0, −3.0
Overall provider rating	46.6	45.5	48.0	−1.9	−4.6, 0.8	−4.4**	−7.0, −1.8
Comprehensiveness	42.6	47.3	43.3	−4.7***	−7.3, −2.1	−0.7	−3.2, 1.8
Coordination	51.5	55.8	56.4	−4.3**	−7.3, −1.2	−4.9**	−7.8, −2.0
Medication shared decision-making	38.2	39.1	39.4	−0.9	−4.0, 2.2	−1.2	−4.2, 1.8
Support for self-management	40.2	44.3	44.5	−4.2**	−6.7, −1.6	−4.3**	−6.8, −1.8

Primary care domain	Probability of negative experience			Comparison: OUD minus other SUD		Comparison: OUD minus no SUD	
	OUD	Other SUD	No SUD	aRD	95% CI	aRD	95% CI
Access	12.9	9.2	8.9	3.7***	2.1, 5.4	4.0***	2.3, 5.6
Communication	15.9	13.0	12.2	2.9**	1.2, 4.6	3.7***	2.0, 5.3
Office Staff	15.3	11.3	11.9	4.1***	2.3, 5.8	3.5***	1.8, 5.2
Overall provider rating	14.4	11.2	10.5	3.2***	1.5, 4.9	3.9***	2.2, 5.5
Comprehensiveness	21.0	16.9	19.5	4.1***	1.9, 6.3	1.5	−0.6, 3.6
Coordination	13.0	11.1	9.5	1.9	−0.2, 4.0	3.5**	1.5, 5.6
Medication shared decision-making	15.0	13.6	13.0	1.4	−0.7, 3.5	2.0	0.0, 4.1
Support for self-management	35.1	28.9	29.7	6.2***	3.6, 8.8	5.4***	2.9, 8.0

aRD: Adjusted risk differences are the absolute difference in the probability of reporting positive or negative healthcare experiences (0 to 100 scale), calculated for patients diagnosed with OUD compared to those with other SUDs and those with no SUD. Estimates were obtained from multivariable multinomial logistic regression models predicting positive, moderate, and negative experiences with care. The models, run separately for each domain, controlled for age, sex, race/ethnicity, education, marital status, U.S. urban vs. rural residence, housing, general health and mental health ratings, Gagne comorbidity index, mental health diagnoses, past-year primary care visits, facility type (medical center versus outpatient clinic), and VHA service region.
A negative RD in positive experiences (top rows) indicate that patients diagnosed with OUD are *less* likely than other patients to report positive healthcare experiences in the domain being modeled. A positive RD in negative experiences (bottom rows) indicate that patients diagnosed with OUD are *more* likely than other patients to report negative healthcare experiences in the domain being modeled.
* $p < .05$.
** $p < .01$.
*** $p < .001$.

buprenorphine to report positive experiences with comprehensiveness (aRD = 8.3) and support for self-management (aRD = 7.1). They reported fewer negative experiences with care coordination (aRD = −4.9) and medication shared decision-making (aRD = −5.4).

3.5. Sensitivity analyses considering the primary care experiences of patients with OUD receiving buprenorphine, naltrexone, and/or methadone treatment

The patterns of differences in experiences observed for patients treated versus not treated with buprenorphine changed when we compared the experiences of patients treated versus not treated with either buprenorphine or naltrexone, and the experiences of patients treated versus not treated with any medication treatment (i.e., buprenorphine, naltrexone, or methadone) for OUD (see Appendix E). In the domains of comprehensiveness, care coordination, medication decision-making, and support for self-management, the aRD estimates were similar or somewhat smaller when the treatment comparison was buprenorphine/naltrexone versus no buprenorphine/naltrexone, than when the comparison was just buprenorphine versus no buprenorphine (Appendix E, Comparison 1 versus 2). In these same domains, the aRD estimates associated with any medication treatment (i.e., buprenorphine, naltrexone, or methadone) were much smaller than the aRD estimates associated with just buprenorphine (Appendix E, Comparison 3 versus 1).

4. Discussion

4.1. Summary of findings

While OUD can be effectively managed in primary care, the experiences of patients with OUD in this setting are not well known. On a validated survey of healthcare experiences, patients with OUD reported fewer positive and more negative experiences with care than did other primary care patients, including those with other SUDs. We found some evidence that OUD-diagnosed patients treated with buprenorphine, compared to those not treated with buprenorphine, had more positive perceptions of provider attention to mental health/substance use concerns (i.e., comprehensiveness) and perceptions that providers gave patients the tools to manage their own health (i.e., support for self-management); as well as fewer negative experiences with care coordination and medication shared decision-making. Because patient

Table 4
Adjusted difference in positive and negative primary care experiences for patients with opioid use disorder who were treated with buprenorphine, compared to patients not treated with buprenorphine.

Primary care domain	Positive experiences		Negative experiences	
	aRD	95% CI	aRD	95% CI
Access	−2.4	−7.9, 3.2	−0.1	−4.6, 4.6
Communication	0.6	−7.9, 4.5	−4.1	−9.0, 0.8
Office staff	−1.7	−8.2, 4.3	−2.6	−7.5, 2.3
Overall provider rating	2.3	−3.7, 8.2	−3.2	−7.9, 1.6
Comprehensiveness	8.3**	2.5, 14.2	−1.7	−6.1, 2.8
Care coordination	6.7	−0.1, 13.6	−4.9*	−9.5, −0.3
Medication shared decision making	3.0	−4.2, 10.3	−5.4*	−10.5, −0.4
Support for self-management	7.1*	1.2, 12.9	−2.4	−8.0, 3.2

aRD: Adjusted risk differences are the absolute difference in the probability of reporting negative or positive healthcare experiences (0 to 100 scale) for patients diagnosed with opioid use disorder and treated with buprenorphine (n = 744), compared to diagnosed patients not treated with buprenorphine (n = 2808). Estimates were obtained from multivariable multinomial logistic regression models predicting positive, moderate, and negative experiences with care. The models, run separately for each domain, controlled for age, sex, race/ethnicity, education, marital status, U.S. urban vs. rural residence, housing, general health and mental health ratings, Gagne comorbidity index, mental health diagnoses, past-year primary care visits, facility type (medical center versus outpatient clinic), and VHA service region.
A positive RD in positive experiences (left column) indicates that patients treated with buprenorphine are *more* likely than are patients without buprenorphine to report positive healthcare experiences in the domain being modeled. A negative RD in negative experiences (right column) indicates that patients treated with buprenorphine are *less* likely than are patients without buprenorphine to report negative healthcare experiences in the domain being modeled.
* $p < .05$.
** $p < .01$.

experiences relate to patient and system level outcomes, our findings have implications for initiatives to engage OUD patients in treatment in primary care settings.

4.2. Prior research

The results from this study both corroborate prior research and contribute new findings about the unique experiences of patients with OUD. Our finding that patients with OUD had fewer positive and more negative experiences than patients without any SUD are similar to results of prior studies that have compared the experiences of patients with SUDs to those of patients without any SUD (Burnett-Zeigler et al., 2011; Desai et al., 2005). In addition, our finding that patients with OUD and other SUDs reported the fewest positive experiences with access are partly consistent with VHA access challenges publicized in the media in 2014 (Hicks, 2014), and with results from prior studies documenting dissatisfaction with access to VHA care (Jones et al., 2016; Zickmund et al., 2018). To our knowledge, this is a first national study to report on the primary care experiences of patients with OUD, specifically.

Patients with OUD were 3–7 percentage points less likely than patients with other SUDs to report positive care experiences—a magnitude of difference that is larger than estimates from other studies of racial/ethnic and homeless differences in primary care experiences (Jones et al., 2016; Jones et al., 2017). In addition, we found OUD-associated differences in primary care experiences remained statistically significant after controlling for race/ethnicity, homelessness, mental health diagnoses and other factors that are often associated with poor ratings of care. To inform health disparity interventions, a future line of inquiry could explore the intersection of these vulnerabilities to determine if racial/ethnic minorities, homeless veterans, or other subgroups of patients with OUD are particularly at risk for poor primary care experiences and outcomes. Our finding that patients with OUD had fewer positive and more negative experiences with access and office staff, when compared to patients with other drug use disorders, points toward distinct barriers to care for this population (Lagisetty et al., 2019).

4.3. Interpretation of findings

Many factors may help to account for our finding of less optimal primary care experiences for patients with OUD compared to those of patients with other SUDs. It is possible that patient experiences with pain and disputes with providers over medication treatment to address pain may erode trust and communication between providers and their patients (Matthias et al., 2013; Ruben, Meterko, & Bokhour, 2018). Negative attitudes toward patients with OUD may also prevent providers from establishing trusting relationships with patients (Kennedy-Hendricks et al., 2016), or from communicating in ways that express respect for patients' needs and preferences (van Boekel, Brouwers, van Weeghel, & Garretsen, 2013). In addition, primary care clinics often lack the resources to respond to the complex needs of patients with OUD, and clinicians themselves can lack training or resources in this area (DeFlavio, Rolin, Nordstrom, & Kazal, 2015; Hutchinson, Catlin, Andrilla, Baldwin, & Rosenblatt, 2014).

Although our analyses of primary care experiences for patients treated with buprenorphine were exploratory, the observed patterns of results are compelling. Patients treated with buprenorphine had favorable impressions of care in select primary care domains that reflect priorities of the patient-centered medical home model (e.g., involvement of patient in medication decisions, comprehensive care, coordination with specialists) and have been associated with depression treatment initiation and continuation (Jones et al., 2015; Jones et al., 2018). Thus, it is plausible that positive primary care experiences in

these domains, in particular, could relate to medication treatment for OUD. Alternatively, the findings could reflect a selection bias if, for instance, providers selectively prescribe buprenorphine to patients with favorable care encounters. However, one would expect such bias to result in positive care experiences in all domains; buprenorphine was not associated with patient perceptions of primary care access, provider communication, overall provider ratings, or office staff helpfulness/courtesy. It is important to note that the patterns of more positive and fewer negative experiences with care associated with buprenorphine largely evaporated when the treatment group included methadone—a treatment only available in specialty treatment settings. Thus, any primary care benefit associated with medication treatment was limited to plausible domains and to medications that could be managed in a primary care setting.

4.4. Implications

Of all primary care domains, patients with OUD reported the fewest positive experiences with access, involvement in medication shared-decision making, and overall provider ratings. The importance of these findings are underscored by research from general outpatient populations demonstrating links between positive provider ratings and patient involvement in shared decision-making on medication adherence and treatment outcomes in medical, mental health, and addiction care domains (Anhang Price et al., 2014; Bauer et al., 2014). While patients with OUD tended to report poorer experiences with primary care than did patients with other SUDs, patients treated with buprenorphine had improved perceptions of medication shared decision-making and comprehensiveness relative to patients diagnosed with OUD who do not receive buprenorphine, which could in turn be linked with improvements in treatment retention. While the current study did not examine OUD treatment outcomes, our results provide motivation for future research to examine the patient and system-level benefits associated with positive primary care experiences for this population.

Our finding that buprenorphine treatment did not predict patient experiences with primary care access, provider communication, or office staff helpfulness/courtesy suggests that strategies to improve primary care experiences for patients with OUD should extend beyond addiction treatment options, which are nonetheless important (Abraham, Andrews, Harris, & Friedmann, 2020; Blanco & Volkow, 2019). Service interventions may need to include tangible support for providers and staff who see patients with addiction; communication skills training (Haverfield et al., 2020); and cultivating organizational identity in which clinicians can share with each other the stress they may feel in addressing difficult psychosocial problems in a nurturing and supportive way. In care of homeless populations, who may overlap with OUD, patient experience is enhanced in part through service designs that integrate primary care and other mental and social services, expand time in the clinical encounter, and foster clinical team identity (Jones et al., 2019; Kertesz et al., 2013; O'Toole et al., 2018). Clinics tailored for persons with OUD are underway in one large VHA medical center, and in some Medicaid medical home programs (Clemans-Cope et al., 2017). Further research could examine whether improving practitioner performance in these domains correlates with improvements of patients' perspectives of primary care.

4.5. Limitations

Our categorization of SUD groups is based on administrative records and may be subject to misclassification. Due to the frequent co-occurrence of SUDs, we were unable to compare the perspectives of patients with specific SUDs (e.g., OUD vs. cocaine use disorder, OUD versus cannabis use disorder, etc.). Second, the analyses were limited to diagnosed VHA patients and the results may not generalize to undiagnosed patients or to patients receiving primary care outside the VHA. Third, we observed modest response rates for all groups and the lowest response rate for patients with OUD. Our analyses included survey weights designed to address unequal probabilities of selection and survey nonresponse related to age and gender. Our finding of lower survey response for patients who were nonmarried and those with recent experiences of homelessness and mental health diagnoses suggests the study patterns may not generalize to highly vulnerable patients.

Finally, we were unable to determine care outside the VHA, reasons for primary care visits (e.g., OUD versus other reasons), and whether buprenorphine was prescribed in primary care versus specialty mental health settings, all limiting our ability to draw inferences about the role of primary care in OUD treatment. Despite these limitations, we provided novel insights into the perspectives of patients with OUD in primary care—an important contribution given that patients with OUD are a vulnerable population for whom primary care experiences and outcomes are relatively unknown. We are confident that patients were reporting on their VHA primary care experiences, given that the PCMH-SHEP survey directs patients to think about experiences with their primary care provider. More than half of patients diagnosed with OUD reported 3 or more visits with their provider in the past year, suggesting high engagement with VHA primary care. Furthermore, prior studies have found veterans using VHA and non-VHA services receive a majority of their primary care (> 70% of visits) in VHA facilities (Liu et al., 2011; Yoon, Vanneman, Dally, Trivedi, & Phibbs, 2019). In the case of buprenorphine, our findings highlight a need to determine the perspectives of patients being managed with buprenorphine in primary care, specifically, and to determine if the provision of buprenorphine in primary care enhances OUD-diagnosed patient experiences in other domains, such as provider communication.

5. Conclusions

This study provides a first look at the primary care experiences of VHA patients diagnosed with OUD in a large, nationally representative sample. We identified suboptimal primary care experiences for patients with and without SUDs, and relatively poor experiences for patients with OUD in several primary care domains. As stakeholders encourage more primary care providers to manage OUD, healthcare systems will need to attend to these patients' experiences. Strategies to address suboptimal healthcare experiences need to ensure that any expected benefit of ongoing efforts to manage OUD and to prescribe buprenorphine treatment in primary care is realized.

CRediT authorship contribution statement

Audrey L. Jones: Conceptualization, Formal analysis, Writing - original draft. **Stefan G. Kertesz:** Writing - review & editing. **Leslie R.M. Hausmann:** Methodology, Writing - review & editing. **Maria K. Mor:** Methodology, Writing - review & editing. **Ying Suo:** Data curation, Software. **Warren B.P.Pettey:** Writing - review & editing. **James H. Schaefer:** Resources. **Adi V. Gundlapalli:** Writing - review & editing. **Adam J. Gordon:** Conceptualization, Writing - review & editing, Supervision.

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Declaration of competing interest

Dr. Stefan G. Kertesz asserts past ownership of stock in Merck and Abbot in his personal portfolio, never amounting to > 3% of assets, and sold in December of 2017. He reports no other history of grants, contracts, honoraria, or any other form of income or assets related to the pharmaceutical industry. No other potential conflicts of interest were identified.

Appendix A. Description of patient experience domains, response options, and classification of negative, moderate and positive experiences

Domain: survey items	Response options	Negative	Moderate	Positive
Access: In the last 12 months …	Never, sometimes usually, always (1–4)	1.00–1.99	2.00–3.74	3.75–4.00
…when you phoned this provider's office to get an appointment for care you needed right away, how often did you get an appointment as soon as you needed?				
…when you made an appointment for a check-up or routine care with this provider, how often did you get an appointment as soon as you needed?				
…when you phoned this provider's office during regular office hours, how often did you get an answer to your medical question that same day?				
Communication: In the last 12 months, how often did this provider…	Never, sometimes usually, always (1–4)	1.00–2.99	3.00–3.82	3.83–4.00
…explain things in a way that was easy to understand?				
…listen carefully to you?				
…show respect for what you had to say?				
…spend enough time with you?				
Office staff: In the last 12 months, how often …	Never, sometimes usually, always (1–4)	1.00–2.99	2.50–3.99	4.00
…were clerks and receptionists at this provider's office as helpful as you thought they should be?				
…did clerks and receptionists at this provider's office treat you with courtesy and respect?				
Provider rating:	0–10	0–5	6–9	10
Using any number from 0 to 10, where 0 is the worst provider possible and 10 is the best provider possible, what number would you use to rate this provider?				
Comprehensiveness: In the last 12 months, did anyone in this provider's office …	No, yes (0,1)	0	0.01–0.99	1.0
…ask you if there was a period of time when you felt sad, empty, or depressed?				
…talk about things in your life that worry you or cause you stress?				
…talk about a personal problem, family problem, alcohol use, drug use, or a mental or emotional illness?				
Care coordination: In the last 12 months, how often did [provider] seem informed and up-to-date about the care you got from specialists?	Never, sometimes usually, always (1–4)	1	2–3	4
Medication shared decision-making: When you talked about starting or stopping a prescription medicine …	Never, sometimes usually, always (1–4)	1.00–2.00	2.01–3.67	3.68–4.00
…how much did this provider talk about the reasons you might want to take the medicine?				
…how much did this provider talk about the reasons you might not want to take a medicine?				
…did this provider ask you what you thought was best for you?	No, yes (1,4)			
Support for self-management: In the last 12 months, did anyone in this provider's office …	No, yes (0,1)	0	0.01–0.99	1.00
…talk with you about specific goals for your health?				
…ask you if there are things that make it hard for you to take care of your health?				

Domains were drawn from the Veterans Health Administration (VHA)'s Patient-Centered Medical Home Survey of Healthcare Experiences of Patients, 2013–2015. Domain scores were calculated as the average of applicable, nonmissing items within each domain, and then categorized as negative, moderate, or positive using cutoffs published in prior studies.

Appendix B. Characteristics of survey respondents and non-respondents, by type of substance use disorder diagnosis

	Opioid use disorder		
	Respondents	Non-respondents	Total
	n = 3939	n = 16,815	n = 20,754
Age 55 and older	71.2%	44.5%	49.6%
Female sex	5.7%	7.5%	7.2%
Married	33.6%	26.5%	27.8%
Any mental health diagnosis	77.3%	81.9%	81.0%
Homeless experience	29.9%	43.9%	41.2%
	Other substance use disorder		
	Respondents	Non-respondents	Total
	n = 40,389	n = 95,037	n = 135,426
Age 55 and older	77.5%	52.4%	59.8%
Female sex	4.3%	6.5%	5.9%
Married	38.2%	29.9%	32.3%
Any mental health diagnosis	71.0%	78.1%	76.0%
Homeless experience	28.1%	31.6%	19.8%
	No substance use disorder		
	Respondents	Non-respondents	Total
	n = 843,550	n = 989,373	n = 1,832,922
Age 55 and older	90.1%	64.0%	76.0%
Female sex	4.8%	9.9%	7.5%
Married	62.5%	51.7%	56.1%
Any mental health diagnosis	21.8%	32.5%	27.6%
Homeless experience	2.3%	5.6%	4.1%

Patients sampled for potential participation in the Veterans Health Administration (VHA)'s Patient-Centered Medical Home Survey of Healthcare Experiences of Patients, 2013–2015. X^2 statistics, run separately for patients in each diagnosis group, were used to test for differences in socio-demographic characteristics of survey respondents compared to non-respondents. All comparisons were statistically significant at $p < .001$.

Appendix C. Differences in positive and negative primary care experiences for patients diagnosed with opioid use disorder, compared to patients with other substance use disorders or no substance use disorder

Primary care domain	Unadjusted percent with positive experience			Comparison: OUD minus other SUD		Comparison: OUD minus no SUD	
	OUD	Other SUD	No SUD	RD	95% CI	RD	95% CI
Access	31.2	36.5	43.4	−5.3***	−7.9, −2.8	−12.2***	−14.7, −9.8
Communication	49.8	55.6	62.8	−5.7***	−8.4, −3.1	−13.0***	−15.5, −10.5
Office Staff	46.6	56.1	59.0	−9.5***	−12.1, −6.8	−12.3***	−14.8, −9.8
Overall provider rating	39.5	43.5	48.2	−4.0***	−6.6, −1.4	−8.7***	−11.2, −6.2
Comprehensiveness	52.8	56.3	42.6	−3.5*	−6.1, −0.8	10.2***	7.7, 12.8
Coordination	48.1	54.4	56.5	−6.3***	−9.4, −3.2	−8.4***	−11.3, −5.5
Medication shared decision-making	38.2	40.2	39.3	−2.0	−5.1, 1.1	−1.1	−4.0, 1.9
Support for self-management	42.8	47.6	44.2	−4.8***	−7.4, −2.1	−1.4	−3.9, 1.1

Primary care domain	Unadjusted percent with negative experience			Comparison: OUD minus other SUD		Comparison: OUD minus no SUD	
	OUD	Other SUD	No SUD	RD	95% CI	RD	95% CI
Access	17.7	11.9	8.7	5.8***	3.6, 8.1	9.0***	6.8, 11.1
Communication	23.4	17.6	11.9	5.8***	3.5, 8.1	11.6***	9.4, 13.8
Office Staff	21.1	14.5	11.6	6.6***	4.4, 8.9	9.5***	7.3, 11.7
Overall provider rating	20.6	14.7	10.2	5.9**	3.7, 8.2	10.3***	8.1, 12.5

Comprehensiveness	19.1	15.5	19.6	3.4***	1.6, 5.6	−0.5	−2.4, 1.4
Coordination	15.1	11.8	9.5	3.3***	0.8, 5.8	5.7***	3.3, 8.0
Medication shared decision-making	18.1	15.0	12.9	3.1***	0.7, 5.5	5.2***	2.9, 7.5
Support for self-management	34.6	27.5	29.8	7.1***	4.5, 9.7	4.8***	2.4, 7.3

* $p < .05$.
** $p < .01$.
*** $p < .001$.

RD: Risk differences are the absolute difference in the percentage of patients reporting positive or negative healthcare experiences (0 to 100 scale), calculated for patients diagnosed with OUD compared to those with other SUDs and those with no SUD. Estimates were obtained from multinomial logistic regression models predicting positive, moderate, and negative experiences with care. A negative RD in positive experiences (top rows) indicate that patients diagnosed with OUD are *less* likely than other patients to report positive healthcare experiences in the domain being modeled. A positive RD in negative experiences (bottom rows) indicate that patients diagnosed with OUD are *more* likely than other patients to report negative healthcare experiences in the domain being modeled

Appendix D

Adjusted probability of reporting positive primary care experiences for patients diagnosed with opioid use disorder, compared to patients with alcohol use disorder, other drug use disorders or no substance use disorders.

Domain	Diagnosis group			
	OUD n = 3544	DUD n = 11,911	AUD n = 24,264	No SUD n = 753,386
Access	39.5[a]	40.2[a]	44.3[b]	42.9[c]
Communication	57.4[a]	58.8[a]	62.4[b]	62.4[b]
Office staff	53.1[a]	58.9[b]	60.9[c]	58.6[b]
Overall provider rating	43.5[a]	43.8[a]	46.6[b]	48.1[c]
Comprehensiveness	42.4[a]	42.9[a]	50.0[b]	43.3[a]
Care coordination	51.4[a]	53.5[a]	57.2[b]	56.4[b]
Medication shared decision making	38.1[a,b]	37.2[b]	40.3[a]	39.4[a]
Support for self-management	40.1[a]	41.8[a]	45.9[b]	44.5[c]

OUD = any opioid use disorder diagnosis; DUD = drug use disorder diagnosis, including those related to amphetamines, cannabis, cocaine, hallucinogens, sedatives, other substances, and unspecified combinations of drugs; AUD = alcohol use disorder; No SUD = patients lacking a diagnosis of OUD, DUD or AUD.

Patients with more than one type of diagnosis were coded in the highest level (i.e., left-most) category. For example, persons with both OUD and DUD diagnoses were coded as OUD; those with DUD and AUD diagnoses were categorized as DUD. Predicted probabilities (reported as percentages, 0–100 scale) and differences in probabilities (risk differences) were estimated from multivariable multinomial logistic regression models predicting positive, moderate, or negative experiences with care. The models, run separately for each domain, controlled for age, sex, race/ethnicity, education, marital status, U.S. urban vs. rural residence, housing, general health and mental health ratings, Gagne comorbidity index, mental health diagnoses, past-year primary care visits, facility type (medical center versus outpatient clinic), and VHA service region.

Groups with different superscripts are statistically different from each other at $p < .05$. In the domain of access, patients in the OUD and DUD groups share the superscript “a” but not “b” or “c”. Thus, there are no differences in access experiences for patients with OUD, compared to patients with DUD; patients with OUD are less likely than patients with AUD and those with no SUD to report positive experiences with access.

Adjusted probability of reporting negative primary care experiences for patients diagnosed with opioid use disorder, compared to patients with alcohol use disorder, other drug use disorders or no substance use disorders.

Domain	Diagnosis group			
	OUD n = 3544	DUD n = 11,911	AUD n = 24,264	No SUD n = 753,386
Access	12.9[a]	10.2[b]	8.5[c]	8.9[c]
Communication	15.9[a]	14.6[a]	11.9[b]	12.2[b]
Office staff	15.4[a]	12.1[b]	10.7[c]	11.9[b]
Overall provider rating	14.5[a]	12.8[a]	10.1[b]	10.5[b]
Comprehensiveness	21.1[a]	19.3[a]	15.4[b]	19.6[a]
Care coordination	13.1[a]	12.4[a]	10.3[b]	9.5[b]
Medication shared decision making	15.1[a]	15.7[a]	12.3[b]	13.0[b]
Support for self-management	35.3[a]	32.6[a]	26.7[b]	29.7[c]

OUD = any opioid use disorder diagnosis; DUD = drug use disorder diagnosis, including those related to amphetamines, cannabis, cocaine, hallucinogens, sedatives, other substances, and unspecified combinations of drugs; AUD = alcohol use disorder; No SUD = patients lacking a diagnosis of OUD, DUD or AUD.

Patients with more than one type of diagnosis were coded in the highest level (i.e., left-most) category. For example, persons with both OUD and DUD diagnoses were coded as OUD; those with DUD and AUD diagnoses were categorized as DUD. Predicted probabilities (reported as percentages,

0–100 scale) and differences in probabilities (risk differences) were estimated from multivariable multinomial logistic regression models predicting positive, moderate, or negative experiences with care. The models, run separately for each domain, controlled for age, sex, race/ethnicity, education, marital status, U.S. urban vs. rural residence, housing, general health and mental health ratings, Gagne comorbidity index, mental health diagnoses, past-year primary care visits, facility type (medical center versus outpatient clinic), and VHA service region.

Groups with different superscripts are statistically different from each other at $p < .05$. In the domain of communication, patients in the OUD and DUD groups share the superscript "a" but not "b". Thus, there are no differences in communication experiences for patients with OUD, compared to patients with DUD; patients with OUD are more likely than patients with AUD and those with no SUD to report negative experiences with communication.

Appendix E. Sensitivity analysis comparing positive and negative experiences for the following groups of patients with opioid use disorder: (1) buprenorphine treatment versus no buprenorphine, (2) buprenorphine or naltrexone treatment versus no buprenorphine or naltrexone, and (3) any medication treatment versus no medication

	Comparison 1	Comparison 2	Comparison 3
Positive experiences	Buprenorphine versus no buprenorphine	Buprenorphine or naltrexone versus no buprenorphine or naltrexone	Any medication versus no medication
Access	RD = -2.4, p = .405	RD = -2.7, p = .314	RD = -5.4, p = .026
Communication	RD = 0.6, p = .832	RD = -0.4, p = .905	RD = −3.2, p = .217
Office	RD = -1.7, p = .593	RD = -1.3, p = .661	RD = -5.5, p = .034
Provider	RD = 2.3, p = .459	RD = 0.6, p = .836	RD = -3.0, p = .236
Comprehensiveness	RD = 8.3, p = .005	RD = 8.5, p = .003	RD = 5.5, p = .032
Coordination	RD = 6.7, p = .055	RD = 6.6, p = .051	RD = 1.5, p = .620
Decision-making	RD = 3.0, p = .413	RD = 1.0, p = .778	RD = -3.2, p = .281
Self-management	RD = 7.1, p = .018	RD = 4.3, p = .138	RD = 2.5, p = .319
Negative experiences	Buprenorphine versus no buprenorphine	Buprenorphine or naltrexone versus no buprenorphine or naltrexone	Any medication versus no medication
Access	RD = -0.1, p = .987	RD = 0.6, p = .779	RD = 3.0, p = .145
Communication	RD = -4.1, p = .101	RD = -2.6, p = .305	RD = -2.3, p = .307
Office	RD = -2.6, p = .295	RD = -2.9, p = .215	RD = -0.6, p = .799
Provider	RD = -3.2, p = .193	RD = -2.0, p = .397	RD = -1.3, p = .554
Comprehensiveness	RD = -1.7, p = .462	RD = -2.4, p = .264	RD = -0.1, p = .948
Coordination	RD = -4.9, p = .038	RD = -4.4, p = .060	RD = -3.5, p = .106
Decision-making	RD = -5.4, p = .034	RD = -3.9, p = .137	RD = -2.2, p = .341
Self-management	RD = -2.4, p = .404	RD = 0.2, p = .946	RD = 0.3, p = .908

Adjusted risk differences are the absolute difference in the probability of reporting positive or negative healthcare experiences (0 to 100 scale), calculated for patients in the treatment group compared to patients not in the defined treatment group. Any medication treatment includes buprenorphine, naltrexone, or methadone. Estimates were obtained from multivariable multinomial logistic regression models predicting positive, moderate, and negative experiences with care. The models, run separately for each domain, controlled for age, sex, race/ethnicity, education, marital status, U.S. urban vs. rural residence, housing, general health and mental health ratings, Gagne comorbidity index, mental health diagnoses, past-year primary care visits, facility type (medical center versus outpatient clinic), and VHA service region.

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